

Tip #2: Get trained to have an extra edge

If you've just entered the college, your course curriculum may not provide all the learning you need. Make an effort to gain some extra skill, be it technical or interpersonal.

"Each engineering institute or college should make use of the vast resource of experts and organisations readily available in the country to do some technical training on specialised topics. This will enable the students for the working world," suggests Shetty.

Electronics has many branches today, such as VLSI, DSP, communications, power electronics and embedded systems. "The industry may assume that the student has been exposed to at least one course in these branches. While the industry may be willing to provide job-specific training, the student is expected to have the technical know-how required to absorb the training. For example, for training in VLSI, the student is expected to know about MOS transistor operation, CMOS circuits, logic gates and flip-flops, operational amplifiers, feedback amplifiers, poles and zeroes," says Ravikumar.

Ravikumar also feels that system-level design using off-the-shelf ICs is a major gap-area today. In digital and analogue design lab classes, students use simple ICs such as gates and op-amps. They may later do a lab on microprocessor/microcontroller/DSP/C programming. Here learning could be improved in two ways:

1. The lab exercises in these classes are often 'canned' — there are ready-made kits with standard set of experiments that the students carry out in a routine fashion. Also, due to large team sizes, all students may not get the required exposure to equipment. However, with falling costs of electronics, students may be able to set up their own little lab in their hostel room. Availability of public-domain software and low-cost computing platforms has helped the students significantly.

2. There is no class where the learning from the courses in digital, analogue, processor and programming are brought together. This may happen in a project for some students.

According to Ahuja, "India produces a large number of electronics and computer science graduate engineers every year. So while there is no dearth of manpower, the challenge is finding 'design-aware' engineers who are trained specifically in VLSI design and can ramp up quickly."

"With the surge in high-tech design projects coming to India in areas such as telecom, automotive, aerospace and industrial automation, the industry workforce has been exposed to very high levels of product design, development, testing and validation phases over the last decade or so. With more and more such projects becoming a norm in India, the industry requires talent with quality and specialised skill sets. The demand for electronics design engineers having product, domain and software tools expertise is high," says Varma.

Kant suggests, "Actively look out for science, engi-



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